Aleksandar Avdalović

Ph.D. Student in Computer Science, Louisiana State University

LinkedIn | Email | Google Scholar | Personal Website | GitHub

Research Interests: Trustworthy and interpretable AI for cyber-physical security and safety, from industrial control systems and power grids to vehicle networks.
Methods: Neurosymbolic AI, symbolic regression, uncertainty quantification through conformal prediction, topological deep learning, formal verification.

Education

Louisiana State University *Baton Rouge, LA, USA*
Ph.D. in Computer Science Aug. 2023 – Jul. 2027 (expected)
GPA: **3.85**
Advisor: Dr. Elias Bou-Harb Committee: Dr. Xugui Zhou, Dr. Dong Lao

University of Primorska, FAMNIT *Koper, Slovenia*
M.Sc. in Data Science Oct. 2020 – Aug. 2023
GPA: **3.89** (U.S. equivalent, ECE evaluation)
Thesis: *Translation of Random Forests into Loop-Free Imperative Programs for Formal Verification*
Advisor: Dr. Raul Barbosa (University of Coimbra)

University of Primorska, FAMNIT *Koper, Slovenia*
B.Sc. in Mathematics Oct. 2017 – Jul. 2020
GPA: **4.00** (U.S. equivalent, ECE evaluation)

Additional Academic Study

University of Coimbra *Coimbra, Portugal*
Master's thesis research visit, Centre for Informatics and Systems (CISUC) Feb. 2022 – Jun. 2022

Eötvös Loránd University (ELTE) *Budapest, Hungary*
ERASMUS+ Exchange Student, Applied Mathematics Feb. 2021 – Jun. 2021

Publications

Conference Proceedings

1. A. N. Massaad, **Aleksandar Avdalović**, P. Andras, J. Khoury, and E. Bou-Harb. *CANDLe: Discerning Deep Learning Methods for CAN Bus Intrusion Detection via a Reproducible and Extensible Benchmarking Framework.* 29th International Symposium on Research in Attacks, Intrusions and Defenses (RAID), 2026. [code]

2. **Aleksandar Avdalović**, J. Khoury, and E. Bou-Harb. *GRACE-ICS: Graph-Aware Conformal Intervals for Industrial Operations Assurance.* IEEE/IFIP Network Operations and Management Symposium (NOMS), 2026, pp. 1–8. [doi] [code]

3. A. N. Massaad*, **Aleksandar Avdalović***, P. Andras, J. Khoury, and E. Bou-Harb. *Beyond Arbitrary Thresholds: Conformal Prediction for Trustworthy CAN Bus Intrusion Detection.* IEEE International Conference on Communications (ICC), 2026, pp. 1–6. [doi]
**Equal contribution.*

4. **Aleksandar Avdalović**, J. Khoury, A. F. Taha, and E. Bou-Harb. *Enhancing Network Security Management in Water Systems using FM-based Attack Attribution.* IEEE/IFIP Network Operations and Management Symposium (NOMS), 2025, pp. 1–10. [doi] [arXiv]

5. I. T. Leong, **Aleksandar Avdalović**, and R. Barbosa. *Program Verification for Rigorous Analysis of Decision Tree Ensembles Against Specifications.* IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W), 2025, pp. 55–62. [doi] [code]

Journal Articles

1. **Aleksandar Avdalović**, J. Khoury, I. AlJabea, M. Hajij, and E. Bou-Harb. *Beyond Flat Models: Preserving ICS Topology using Combinatorial Complexes for Hierarchy-Aware Intrusion Detection.* IEEE Transactions on Network and Service Management (TNSM), 2026, early access. [doi]

Papers Under Review and Manuscripts

1. **Aleksandar Avdalović** and E. Bou-Harb. *Follow the Power: Symbolic and Readable Intrusion Detection for Reconfigurable Power Networks.* IEEE International Conference on Communications (ICC), 2027. [code]
2. **Aleksandar Avdalović**, J. Khoury, and E. Bou-Harb. *Calibrating the Alarm: Why Deep Learning ICS Anomaly Detectors Fail at Deployment.* IEEE Transactions on Network and Service Management. ***Major revision.*** [code]
3. **Aleksandar Avdalović**, A. N. Massaad, J. Khoury, and E. Bou-Harb. *ACID: Automatic and Comprehensible Intrusion Detection for Industrial Control Systems via Symbolic Regression.* To be submitted to IEEE/IFIP DSN 2027.

Teaching & Mentorship

Graduate Research Mentor — Louisiana State University
Cyber Threat Intelligence Lab — 2025 – Present

- Mentor a junior Ph.D. student on experimental design and evaluation rigor, and on bringing uncertainty quantification into vehicular network security. The work led to two co-authored papers, at IEEE ICC 2026 and RAID 2026.

Guest Lecturer — Louisiana State University
CSC 2362, Introduction to Cybersecurity — Fall 2026

- Lecture on trustworthy AI for process-level intrusion detection in industrial control systems, covering how symbolic models let engineers check a detector against known physics before deployment.

Student Math Tutor — Slovenia
Secondary school and university students — 2017 – 2022

- Taught calculus, linear algebra, and discrete mathematics in one on one sessions, adapting explanations to each student's background.

Courses Ready to Teach

Undergraduate. Introduction to Cybersecurity, Introduction to Machine Learning, Fundamentals of Artificial Intelligence, Deep Learning, Data Structures and Algorithms, Database Systems, Discrete Mathematics, Probability and Statistics, Linear Algebra.

Graduate. Network and Systems Security, Statistical Machine Learning, Applied Deep Learning, Advanced Algorithms, Formal Verification, Uncertainty Quantification, Geometric and Topological Deep Learning, Multivariate Statistics.

Proposed new course. *Trustworthy AI for Cyber-Physical Security and Safety.* A graduate seminar on what guarantees a model must provide before its output can guide a physical action, combining geometric deep learning, uncertainty quantification, and state-space models of physical processes.

Appointments

Graduate Research Assistant, Cyber Threat Intelligence Lab Aug. 2023 – Present
Louisiana State University, Baton Rouge, LA
Advisor: Dr. Elias Bou-Harb

- Study what a deep learning model must demonstrate before its output can guide a decision in a running cyber-physical system (CPS), and build systems that meet that standard. Industrial control systems (ICS) and vehicle networks are the main settings.
- Show that deep learning detectors in CPS are optimistically evaluated and lose their usefulness once deployed, raising false alarms so often that operators cannot act on them. Develop adaptive conformal calibration that brings the false alarm rate back to the level the detector was calibrated for, without retraining it.
- Build ICS detectors on the defender's knowledge of the plant. Topological neural networks encode how the system is organized (Purdue model), and symbolic regression learns its physics from normal operation as equations an engineer can check, including power balance equations that stay valid as operators reconfigure a power grid.

Visiting Researcher, Software and Systems Engineering Group Feb. 2022 – Jun. 2022
CISUC, University of Coimbra, Portugal

- Translated random forest models into loop-free imperative programs for formal verification in safety-critical settings, published at IEEE/IFIP DSN-W 2025.

Research Presentations

- **IEEE/IFIP NOMS 2026** Rome, Italy
  Oral Presentation; *GRACE-ICS: Graph-Aware Conformal Intervals for Industrial Operations Assurance* May 2026
- **IEEE/IFIP NOMS 2025** Honolulu, HI, USA
  Oral Presentation; *Enhancing Network Security Management in Water Systems using FM-based Attack Attribution* May 2025

Schools & Symposia Attended

- **15th Annual Machine Learning Symposium** New York, NY, USA
  Participant Oct. 2024
- **9th PhD Summer School in Discrete Mathematics** Rogla, Slovenia
  Participant Jul. 2019

Professional Service

- Session Chair, Experience Session, IEEE/IFIP Network Operations and Management Symposium (NOMS), Rome, Italy, 2026.
- Reviewer, IEEE Transactions on Network and Service Management (TNSM).

Grants & Scholarships

- **$25,900** | University of Primorska Scholarship for Outstanding Achievements, Koper, Slovenia, 2017–2022.
- **$2,700** | ERASMUS+ Thesis Research Scholarship, University of Coimbra, Portugal, 2022.

- **$2,350** | ERASMUS+ Exchange Scholarship, Eötvös Loránd University, Budapest, Hungary, 2021.
- **$1,950** | ERASMUS+ Traineeship Scholarship, Vienna, Austria, 2021.
- **$500** | Student Travel Grant, IEEE/IFIP NOMS, Honolulu, HI, USA, 2025.

Amounts awarded in euros are shown in U.S. dollars at the September 2026 exchange rate (1 EUR = 1.15 USD).

Honors & Awards

- Gold Medal, Primatijada Student Mathematical Competition, Golden Sands, Bulgaria, 2019.
- Competitor, Al-Khorezmi International Mathematical Olympiad, Urgench, Uzbekistan, 2018.

Industry Experience

Business Intelligence Consultant / Data Warehouse Engineer Aug. 2022 – Aug. 2023
CRMT, Ljubljana, Slovenia

- Built data warehouse and business intelligence solutions with SQL and ETL pipelines for clients in insurance, retail, and telecommunications.
- Developed applied machine learning models for public sector and telecommunications clients, including a recommender system built on top of the clients' ETL pipelines.

Data Analyst Trainee Jul. 2021 – Sep. 2021
ClearKarma, Vienna, Austria

- Supported market analysis, data extraction, and prototype development for data-driven product features.

Technical Skills

Programming. Python, Java, R, SQL, LaTeX.
Machine learning and verification. PyTorch, scikit-learn, TopoX (TopoNetX, TopoModelX), PySR, OpenJML, CUDA and GPU training.
Data and systems. Git, Linux, ETL pipelines, data warehousing.
Research practice. Reproducible experiment pipelines, benchmarking frameworks, open-source release.
Languages. Serbian (native), English and Slovenian (full professional proficiency).

Professional References

Available upon request.